Exp no. 01

Comparative study of Low-code development versus Traditional application development approaches.

Aim:

To study & compare low-code Development & Traditional application development based on their features, advantages, development process, cost, flexibility & overall efficiency in application development.

Features of Traditional application development Vs Low-code development

Traditional application development	Low-code development
• Requires extensive programming knowledge	• Request little (or) no coding.
• Development is done manually by writing source code	• Uses drag & drop components & visuals interfaces.
• Longer development lifecycle	Faster app development
• Highly customizable & flexible	• Limited customization compared to traditional development
• Requires experienced software developers	• Can be used by both developers & non-developers

• Higher maintance effort	• Easier maintenance with built-in tools
• Suitable for large & complex app	• Best suited for small to medium business app & rapid prototyping
• Manual testing & deployment	Automated testing & one-click deployment features.

Advantages of Traditional development vs Low-code Development

Traditional app Development:

- Provides complete control over app design & functionality
- Suitable for highly complex & enterprise-level app.
- offers better scalability & performance optimization
- Supports unlimited customization
- Easier integration with specialized hardware & software

Low-code development:

- Faster app development & deployment.
- Reduces development cost
- Reduces fewer coding skills.
- Increase productivity via reusable components
- Enables rapid a prototyping & quick modifications.

- Simplifies app maintenance.
- Allows business users to participate in app development.

Comparative Study:

Parameter	Traditional develop.	low-code Development
Coding	Extensive coding req	minimal coding req.
Development speed	slow	Fast
cost	high	low
Learning curve	steep	easy
Flexibility	very high	moderate
Customization	Unlimited	limited
Maintanence	more complex	easier
Scalability	Excellent	Good
Deployment	Manual	Automated

Result:

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The study concludes that low-code development enables faster & cost-effective app dev. with minimal coding. & is greater for complex applications.